Problem

Develop the state equations for the following differential equation.

$$\frac{d^2y(t)}{dt^2} + \frac{6\,dy(t)}{dt} + 7y(t) = z(t)$$

Step-by-step solution

Step 1 of 3

Consider the following differential equation.

$$\frac{d^2y(t)}{dt^2} + 6\frac{dy(t)}{dt} + 5y(t) = z(t) \quad \text{...... (1)}$$

Select the state variables:

Let, $x_1 = y(t)$.

Therefore,

$$\dot{x}_1 = \dot{y}(t) \quad \text{...... (2)}$$

Consider,

$$x_2 = \dot{x}_1 = \dot{y}(t) \quad \text{...... (3)}$$

Now, look at the second order system that would normally have two first-order terms in solution.

$$\dot{x}_2 = \ddot{y}(t)$$

Step 2 of 3

Recall equation (1),

$$\frac{d^2y(t)}{dt^2} + 6\frac{dy(t)}{dt} + 5y(t) = z(t)$$

$$\ddot{y}(t) + 6\dot{y}(t) + 7y(t) = z(t)$$

Substitute $\dot{y}(t)$ for $\dot{x}_1$ and $\ddot{y}(t)$ for $\dot{x}_2$ in the equation.

$$\dot{x}_2 + 6\dot{x}_1 + 7y(t) = z(t)$$

$$\dot{x}_2 = -6\dot{x}_1 - 7y(t) + z(t)$$

Substitute $\dot{x}_1$ for x_2 and $y(t)$ for x_1 in the equation.

$$\dot{x}_2 = -6x_2 - 7x_1 + z(t) \quad \text{...... (4)}$$

Step 3 of 3

Write the state equations from equations (2), (3) and equation (4).

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -7x_1 - 6x_2 + z(t)$$

$$y(t) = x_1$$

Write the state equations in matrix form.

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -7 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} z(t)$$

$$y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + [0] z(t)$$

Thus, the state equations for the given differential equation is,

$$\boxed{\begin{aligned} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} &= \begin{bmatrix} 0 & 1 \\ -7 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} z(t) \\ y(t) &= \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + [0] z(t) \end{aligned}}.$$